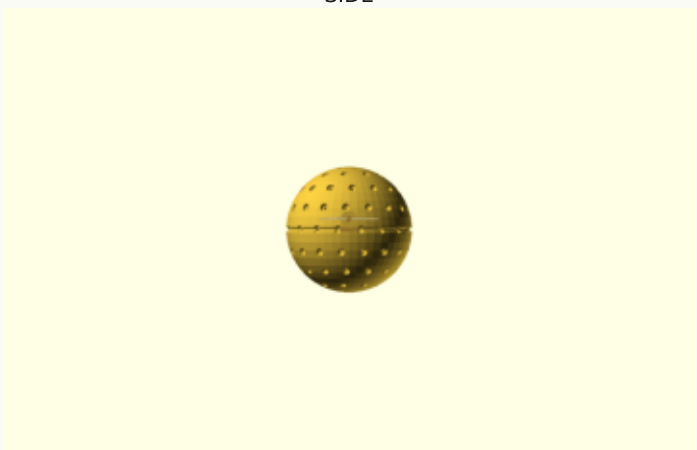


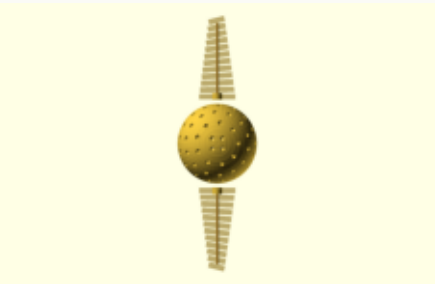
ISOMETRIC



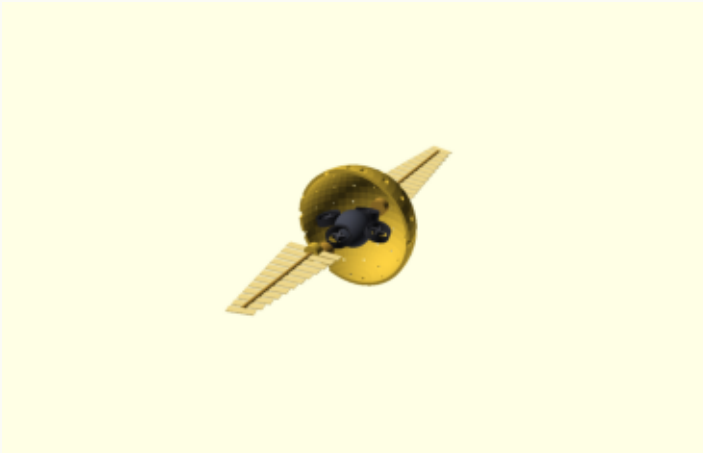
SIDE



TOP



CUTAWAY — internal drive revealed



KEY DIMENSIONS & SPEC

(every number ties back to the flight software)

Shell radius40 mm (Ø80)

ball/params.py snitch.radius = 0.04 m

Mass target50 g

snitch.mass = 0.05 kg

Top speed19 m/s

snitch.max\_speed

Max accel32 m/s²

snitch.max\_accel (absurdly agile)

Lift4 × micro-EDF (internal)

swarm core – the real flight

Wings2 × flapping, 8-14 Hz

coreless gearmotor driven

Capturehand held 0.25 s

capture\_radius 0.22 / capture\_dwll

Danger sense4.0 m

snitch.danger\_radius

NOTE

OFFICIAL STORY: it flies by flapping its wings. The wings are thin and genuinely flutter (SNT-02/03/04). The lift is actually the internal micro-EDF swarm (SNT-05).

Do not explain this to Potterheads.

BILL OF MATERIALS — SNT (14 line items, prototype qty 1)

| REF    | PART                             | SPEC                                          | QTY | \$ EA | \$ EXT |
|--------|----------------------------------|-----------------------------------------------|-----|-------|--------|
| SNT-01 | Gold-clad vented sphere shell    | Ø80 mm (R=40 mm), walnut-sized, dimple vents  | 1   | 90    | 90     |
| SNT-02 | Animatronic flapping wing ass... | thin filigree vane on hinge, span ~76 mm      | 2   | 28    | 56     |
| SNT-03 | Coreless gearmotor + crank       | 6 mm coreless, flap linkage, 8-14 Hz beat     | 2   | 9     | 18     |
| SNT-04 | Wing membrane                    | 5 micron gold-tint mylar, feather-cut         | 2   | 4     | 8      |
| SNT-05 | Micro ducted-fan lift unit       | Ø16 mm internal, the ACTUAL lift              | 4   | 22    | 88     |
| SNT-06 | Micro brushless motor            | 4 mm, paired to micro-EDF                     | 4   | 11    | 44     |
| SNT-07 | Micro ESC (4-in-1)               | 5 A, integrated                               | 1   | 26    | 26     |
| SNT-08 | Nano flight controller           | STM32 class, IMU on-board, predictive evasion | 1   | 38    | 38     |
| SNT-09 | Optical-flow + ToF sensor        | 360deg-ish reach detection                    | 2   | 16    | 32     |
| SNT-10 | Micro LiDAR / proximity ring     | detects incoming hands/airframes              | 1   | 45    | 45     |
| SNT-11 | Capacitive capture sensor        | registers an enclosing hand                   | 1   | 12    | 12     |
| SNT-12 | Micro Li-Po cell                 | 1S 300 mAh, lightweight                       | 1   | 8     | 8      |
| SNT-13 | Gold micro-LED                   | glow so the crowd can track it                | 1   | 5     | 5      |
| SNT-14 | 2.4 GHz telemetry link           | reports pose to the Dementor referee          | 1   | 14    | 14     |

UNIT TOTAL (prototype):\$484